Name: ANSWER KEY **Date:** ________________
Course: CS 471 - Artificial Intelligence
Lesson 15: Reinforcement Learning III (Q-Learning)

Section 1: Instructor-Led Walkthrough

Objective: Calculate a Q-Value update for an autonomous perimeter sentry.

Given the Q-learning update equation:

$$Q_{k+1}(s,a) \leftarrow (1-\alpha)Q_k(s,a) + \alpha\left[R(s,a,s') + \gamma \max_{a'} Q_k(s',a')\right]$$

Assume the sentry is at Checkpoint 1 (s), takes action "Advance" (a), and receives a reward of $R = -2$. It transitions to Checkpoint 2 (s'), which has a maximum future Q-value of 10. Let $\alpha = 0.5$, $\gamma = 1.0$, and the initial $Q(s,a) = 0$.

Step 1: Calculate the sample
$sample = -2 + (1.0 \times 10) = 8$

Step 2: Calculate the new Q-value
$Q(s,a) \leftarrow (0.5 \times 0) + (0.5 \times 8) = 4$

Section 2: Student Practice

Problem 1: Exploration vs. Exploitation
An Electronic Warfare algorithm utilizes an ϵ-greedy strategy to find optimal frequencies for jamming adversary comms. If $\epsilon = 0.1$, explain the behavioral tradeoff the algorithm makes at each time step.

With a probability of 0.9 (90%), the algorithm exploits known data by acting on the current policy to maximize rewards. With a probability of 0.1 (10%), it acts randomly to explore the space and discover potentially better jamming frequencies.

Problem 2: Model-Free Learning

Why does an autonomous drone using Q-Learning not require prior knowledge of the transition function $T(s, a, s')$ to learn an optimal flight path?

Model-free learning directly estimates the values (Q-values) of states through observed samples. By using a running average of experienced rewards and subsequent state values, the drone incorporates the environment's probabilistic nature inherently without needing to store or calculate a complete map of state transitions.